Math 314, Final Exam practice (Spring 2026)

Problem 1. Let

$$\mathbf{u} = \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix} \quad \mathbf{v} = \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix} \quad \mathbf{w} = \begin{bmatrix} -5 \\ -2 \\ 1 \end{bmatrix} \quad \mathbf{z} = \begin{bmatrix} -5 \\ 0 \\ 5 \end{bmatrix} \quad A = \begin{bmatrix} 1 & 0 & -5 \\ -2 & 1 & -2 \\ 1 & 2 & 1 \end{bmatrix} \quad B = \begin{bmatrix} 1 & 0 & -5 & -5 \\ -2 & 1 & -2 & 0 \\ 1 & 2 & 1 & 5 \end{bmatrix}.$$

Note that $\mathbf{u}, \mathbf{v}, \mathbf{w}$ are the columns of A and $\mathbf{u}, \mathbf{v}, \mathbf{w}, \mathbf{z}$ are the columns of B.

For each of the following statements, decide if it is true or false and provide justification.

(a) The set $\{\mathbf{u}, \mathbf{v}, \mathbf{w}\}$ is an orthogonal set.

(b) The set $\{\mathbf{u}, \mathbf{v}, \mathbf{w}\}$ is a linearly independent set.

(c) The set $\{\mathbf{u}, \mathbf{v}, \mathbf{w}\}$ spans $\mathbb{R}^3$.

(d) The set $\{\mathbf{u}, \mathbf{v}, \mathbf{w}\}$ is a basis for $\mathbb{R}^3$.

(e) The set $\{\mathbf{u}, \mathbf{v}, \mathbf{z}\}$ is an orthogonal set.

(f) The set $\{\mathbf{u}, \mathbf{v}, \mathbf{z}\}$ is a linearly independent set.

(g) The set $\{\mathbf{u}, \mathbf{v}, \mathbf{z}\}$ spans $\mathbb{R}^3$.

(h) The set $\{\mathbf{u}, \mathbf{v}, \mathbf{z}\}$ is a basis for $\mathbb{R}^3$.

(i) The set $\{\mathbf{v}, \mathbf{w}, \mathbf{z}\}$ is an orthogonal set.

(j) The set $\{\mathbf{v}, \mathbf{w}, \mathbf{z}\}$ is a linearly independent set.

(k) The set $\{\mathbf{v}, \mathbf{w}, \mathbf{z}\}$ spans $\mathbb{R}^3$

(l) The set $\{\mathbf{v}, \mathbf{w}, \mathbf{z}\}$ is a basis for $\mathbb{R}^3$.

(m) $\mathrm{col}(A) = \mathrm{col}(B)$

(n) $\mathrm{null}(A) = \mathrm{null}(B)$

(o) The equation $x_1\mathbf{u} + x_2\mathbf{v} + x_3\mathbf{w} = 0$ has nontrivial solutions.

(p) The matrix A is invertible.

(q) The linear system $A\mathbf{x} = \mathbf{b}$ is consistent for any choice of $\mathbf{b}$ in $\mathbb{R}^3$.

(r) The orthogonal complement of $W = \mathrm{span}\{\mathbf{v}, \mathbf{z}\}$ is $\mathrm{span}\{\mathbf{u}\}$.

(s) The orthogonal complement of $W = \mathrm{span}\{\mathbf{z}\}$ is a line.

(t) The orthogonal projection of $\mathbf{w}$ onto $\mathrm{span}\{\mathbf{u}, \mathbf{v}\}$ is $\mathbf{0}$.

(u) The distance from $\mathbf{z}$ to $\mathrm{span}\{\mathbf{u}, \mathbf{v}\}$ is the same as the distance from $\mathbf{w}$ to $\mathrm{span}\{\mathbf{u}, \mathbf{v}\}$.

Answer 1.

(a) True, since

$$\begin{aligned}
\mathbf{u}\cdot\mathbf{v} &= 0 + (-2)\cdot 1 + 1\cdot 2 = 0\\
\mathbf{u}\cdot\mathbf{w} &= 1\cdot(-5) + (-2)\cdot(-2) + 1\cdot 1 = -5+4+1 = 0\\
\mathbf{v}\cdot\mathbf{w} &= 0 + 1\cdot(-2) + 2\cdot 1 = -2+2 = 0.
\end{aligned}$$

(b) True, since it is an orthogonal set and all orthogonal sets are linearly independent.

(c) True, since it is a linearly independent set of 3 vectors in $\mathbb{R}^3$.

(d) True, since it is a linearly independent set of 3 vectors in $\mathbb{R}^3$.

(e) False, because

$$\mathbf{v}\cdot\mathbf{z} = 2\cdot 5 = 10 \neq 0.$$

(f) Yes, since

$$\begin{bmatrix} | & | & | \\ \mathbf{u} & \mathbf{v} & \mathbf{z} \\ | & | & | \end{bmatrix} = \begin{bmatrix} 1 & 0 & -5 \\ -2 & 1 & 0 \\ 1 & 2 & 5 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

has a pivot in every column.

(g) True, since

$$\begin{bmatrix} | & | & | \\ \mathbf{u} & \mathbf{v} & \mathbf{z} \\ | & | & | \end{bmatrix} = \begin{bmatrix} 1 & 0 & -5 \\ -2 & 1 & 0 \\ 1 & 2 & 5 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

has a pivot in every row.

Alternatively, because we have a set of 3 vectors in $\mathbb{R}^3$ that we know are linearly independent, and thus they must form a basis for $\mathbb{R}^3$ (and must then span $\mathbb{R}^3$).

(h) True, since

$$\begin{bmatrix} | & | & | \\ \mathbf{u} & \mathbf{v} & \mathbf{z} \\ | & | & | \end{bmatrix} = \begin{bmatrix} 1 & 0 & -5 \\ -2 & 1 & 0 \\ 1 & 2 & 5 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

has a pivot in every row and column.

Alternatively, because we have a set of 3 vectors in $\mathbb{R}^3$ that we know are linearly independent, and thus they must form a basis for $\mathbb{R}^3$.

(i) False, because

$$\mathbf{v}\cdot\mathbf{z} = 2\cdot 5 = 10 \neq 0.$$

(j) False, because

$$\begin{bmatrix} | & | & | \\ \mathbf{v} & \mathbf{w} & \mathbf{z} \\ | & | & | \end{bmatrix} = \begin{bmatrix} 0 & -5 & -5 \\ 1 & -2 & 0 \\ 2 & 1 & 5 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

does not have a pivot in every column.

(k) False, because

$$\begin{bmatrix} | & | & | \\ \mathbf{v} & \mathbf{w} & \mathbf{z} \\ | & | & | \end{bmatrix} = \begin{bmatrix} 0 & -5 & -5 \\ 1 & -2 & 0 \\ 2 & 1 & 5 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

does not have a pivot in every row.

(l) False, because

$$\begin{bmatrix} | & | & | \\ \mathbf{v} & \mathbf{w} & \mathbf{z} \\ | & | & | \end{bmatrix} = \begin{bmatrix} 0 & -5 & -5 \\ 1 & -2 & 0 \\ 2 & 1 & 5 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

is not the 3×3 identity matrix.

Alternatively, note we already explained why this is neither linearly independent nor spans $\mathbb{R}^3$.

(m) True, because we saw that the columns of A span $\mathbb{R}^3$, so $\operatorname{col}(A) = \mathbb{R}^3$. Note that $\operatorname{col}(A) \subseteq \operatorname{col}(B) \subseteq \mathbb{R}^3$, so $\operatorname{col}(B)$ must also be $\mathbb{R}^3$.

(n) False. This does not even make sense: $\operatorname{null}(A)$ is a subspace of $\mathbb{R}^3$ while $\operatorname{null}(B)$ is a subspace of $\mathbb{R}^4$.

(o) False. A nontrivial solution to the equation corresponds to an equation of linear dependence among $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$. But we saw this is a linearly independent set, so there are no nontrivial solutions to this equation.

Alternatively, we can see that

$$A = \begin{bmatrix} | & | & | \\ \mathbf{u} & \mathbf{v} & \mathbf{w} \\ | & | & | \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

which has a pivot in every column.

(p) Yes, since we saw that the columns of A are orthogonal and thus linearly independent.

Alternatively, we can show[1] that

$$A = \begin{bmatrix} | & | & | \\ \mathbf{u} & \mathbf{v} & \mathbf{w} \\ | & | & | \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

(q) True, since we saw above that the columns of A span $\mathbb{R}^3$.

(r) True We saw above that $\mathbf{u} \cdot \mathbf{v} = 0$, and

$$\mathbf{u} \cdot \mathbf{z} = 1 \cdot (-5) + 0 + 1 \cdot 5 = 0.$$

Thus $\mathbf{u}$ is orthogonal to W, and thus $\operatorname{span} \mathbf{u} \subseteq W^\perp$. On the other hand, $\mathbf{v}$ and $\mathbf{z}$ are not multiples of each other, so W has dimension 2. Since W is a subspace of $\mathbb{R}^3$, its orthogonal complement has dimension $3 - 2 = 1$. Therefore, $\operatorname{span} \mathbf{u} = W^\perp$.

[1] You would have to do the calculations for that!

(s) False, since W is a subspace of $\mathbb{R}^3$ with dimension 1, so its orthogonal complement has dimension $3 - 1 = 2$.

(t) Yes! We saw above that $\mathbf{w}$ is orthogonal to both $\mathbf{u}$ and $\mathbf{v}$, so $\mathbf{w}$ is in the orthogonal complement of span $\mathbf{u}, \mathbf{v}$, and thus its projection of span $\mathbf{u}, \mathbf{v}$ is 0.

(u) False.

First, note that $\{\mathbf{u}, \mathbf{v}\}$ is an orthogonal set, so it is an orthogonal basis for $W = \text{span}\,\mathbf{u}, \mathbf{v}$. We saw that the orthogonal projection of $\mathbf{w}$ on $W = \text{span}\,\mathbf{u}, \mathbf{v}$ is 0, so the distance from $\mathbf{w}$ to span $\mathbf{u}, \mathbf{v}$ is

$$\text{dist}_W(\mathbf{w}) = \|\mathbf{w} - 0\| = \|\mathbf{w}\| = \sqrt{(-5)^2 + 0^2 + 5^2} = \sqrt{50} = 5\sqrt{2}.$$

On the other hand, the orthogonal projection of $\mathbf{z}$ onto W is

$$\text{proj}_W(\mathbf{z}) = \frac{\mathbf{z} \cdot \mathbf{u}}{\mathbf{u} \cdot \mathbf{u}}\mathbf{u} + \frac{\mathbf{z} \cdot \mathbf{v}}{\mathbf{v} \cdot \mathbf{v}}\mathbf{v} = \frac{0}{1+4+1}\begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix} + \frac{10}{1+4}\begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 0 \\ 2 \\ 4 \end{bmatrix}.$$

Thus

$$\mathbf{z} - \text{proj}_W(\mathbf{z}) = \begin{bmatrix} -5 \\ 0 \\ 5 \end{bmatrix} - \begin{bmatrix} 0 \\ 2 \\ 4 \end{bmatrix} = \begin{bmatrix} -5 \\ -2 \\ 1 \end{bmatrix}$$

$$\text{dist}_W(\mathbf{z}) = \left\|\mathbf{z} - \text{proj}_W(\mathbf{z})\right\| = \sqrt{25 + 4 + 1} = \sqrt{30}.$$

Problem 2. Suppose that the 3×4 matrix A has rref B, about which we are given some partial information.

$$B = \text{rref}(A) = \begin{bmatrix} 1 & 0 & a & b \\ c & 1 & d & e \\ 0 & f & 0 & 0 \end{bmatrix}.$$

(a) What values of a, b, c, d, e, f are determined by the given information?

(b) If the columns of A are (in order) $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$, and $\mathbf{v}_4$, then what must be true about these four vectors in $\mathbb{R}^3$?

Answer 2.

(a) $c = f = 0$.

(b)

$$\mathbf{v}_3 = a\mathbf{v}_1 + d\mathbf{v}_2 \quad \text{and} \quad \mathbf{v}_4 = b\mathbf{v}_1 + e\mathbf{v}_2.$$

Problem 3. Let $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ be a linearly independent set of vectors in $\mathbb{R}^4$. Is there necessarily a vector $\mathbf{b}$ in $\mathbb{R}^4$ such that $\mathbf{b}$ is not in $\text{span}\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$? Explain your answer.

Answer 3. Yes. Since the set $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ is linearly independent, it can be extended to a basis for $\mathbb{R}^4$. But since $\dim(\mathbb{R}^4) = 4$, that means there exists a vector $\mathbf{b}$ such that $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{b}\}$ is a basis for $\mathbb{R}^4$, and in particular that means that $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{b}\}$ is linearly independent and thus that $\mathbf{b}$ is not in $\operatorname{span}\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$.

Problem 4. Let B be a 3×3 matrix with two eigenvalues. Each eigenspace is 1-dimensional. Is B diagonalizable? Why?

Answer 4. No. For B to be diagonalizable, the sums of the dimensions of the eigenspaces needs to be 3.

Problem 5. Suppose that C is a 7×6 matrix such that the null space of C has dimension 4. Find the rank of C, and carefully explain your answer.

Answer 5. By the Rank–Nullity Theorem,

$$\operatorname{rank}(C) + \dim(\operatorname{null}(C)) = 6.$$

Since $\dim(\operatorname{null}(C)) = 4$, we conclude that

$$\operatorname{rank}(C) = 6 - 4 = 2.$$

Problem 6. Let

$$W = \operatorname{span}\left\{\begin{bmatrix}1\\2\\3\\4\\5\end{bmatrix}, \begin{bmatrix}7\\0\\0\\0\\0\end{bmatrix}\right\}.$$

What is $\dim(W^\perp)$?

Answer 6. The two vectors that span W are not multiples of each other, so they are linearly independent, and thus a basis for W. We conclude that $\dim(W) = 2$. Since W is a subspace of $\mathbb{R}^5$, we conclude that

$$\dim(W^\perp) = 5 - 2 = 3.$$

Alternative solution: Note that $W^\perp$ is the null space of

$$A = \begin{bmatrix}1 & 2 & 3 & 4 & 5\\7 & 0 & 0 & 0 & 0\end{bmatrix} \sim \begin{bmatrix}1 & 0 & 0 & 0 & 0\\0 & 2 & 3 & 4 & 5\end{bmatrix}.$$

Since A has three free variables, we conclude that $\dim(W^\perp) = 3$.

Problem 7. Consider the following vectors in $\mathbb{R}^3$:

$$\mathbf{v}_1 = \begin{bmatrix}0\\1\\2\end{bmatrix} \quad \mathbf{v}_2 = \begin{bmatrix}0\\2\\1\end{bmatrix} \quad \mathbf{w}_1 = \begin{bmatrix}2\\1\\1\end{bmatrix} \quad \mathbf{w}_2 = \begin{bmatrix}2\\2\\2\end{bmatrix}.$$

(a) How can we know without computations that the set $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{w}_1, \mathbf{w}_2\}$ is linearly dependent?

(b) Find the parametric solution for the equation $x_1\mathbf{v}_1 + x_2\mathbf{v}_2 + x_3\mathbf{w}_1 + x_4\mathbf{w}_2 = 0$. You may wish to use the fact that

$$\begin{bmatrix} 0 & 0 & 2 & 2 \\ 1 & 2 & 1 & 2 \\ 2 & 1 & 1 & 2 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 & 1/3 \\ 0 & 1 & 0 & 1/3 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

(c) Let

$$V = \operatorname{span}\{\mathbf{v}_1, \mathbf{v}_2\} \qquad W = \operatorname{span}\{\mathbf{w}_1, \mathbf{w}_2\}.$$

Find a vector $\mathbf{u}$ that is both in V and in W. Explain your answer carefully. (*Hint:* Use part (b).)

Answer 7.

(a) Any set of four vectors in $\mathbb{R}^3$ is linearly dependent because $4 > 3 = \dim(\mathbb{R}^3)$.

(b) The solutions are of the form

$$\begin{cases} x_1 = -\frac{x_4}{3} \\ x_2 = -\frac{x_4}{3} \\ x_3 = -x_4 \\ x_4 \text{ is a free variable} \end{cases} \implies x = \begin{bmatrix} -\frac{x_4}{3} \\ -\frac{x_4}{3} \\ -x_4 \\ x_4 \end{bmatrix} = x_4 \begin{bmatrix} -\frac{1}{3} \\ -\frac{1}{3} \\ -1 \\ 1 \end{bmatrix} \quad \text{where } x_4 \text{ is any real number.}$$

(c) Setting $x_4 = 3$ in (b), we get that

$$x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} -1 \\ -1 \\ -3 \\ 3 \end{bmatrix}$$

is a solution to $x_1v_1 + x_2v_2 + x_3w_1 + x_4w_2 = 0$. More precisely,

$$-v_1 - v_2 - 3w_1 + w_2 = 0 \iff -3w_1 + w_2 = v_1 + v_2.$$

Note that $-3w_1 + w_2$ is in W and $v_1 + v_2$ is in V, so

$$\mathbf{u} = \mathbf{v}_1 + \mathbf{v}_2 = \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix} + \begin{bmatrix} 0 \\ 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 3 \\ 3 \end{bmatrix}$$

is in both V and W.

Problem 8. Consider the subspace of $\mathbb{R}^3$

$$W = \operatorname{span}\left\{\begin{bmatrix}2\\1\\1\end{bmatrix}, \begin{bmatrix}-1\\1\\-2\end{bmatrix}\right\}.$$

(a) Find $\dim(W)$ and say what W is as a geometric object.

(b) Find an orthogonal basis for W.

(c) Find the projection of the vector $\mathbf{v} = \begin{bmatrix}1\\2\\0\end{bmatrix}$ onto W.

(d) Find the distance from the point $(1, 2, 0)$ to the subspace W.

(e) Describe $W^\perp$ by giving a basis for $W^\perp$ and describing what it is as a geometric object.

(f) Find a linear equation for which the solution set is W.

Answer 8.

(a) Since the set

$$\left\{\begin{bmatrix}2\\1\\1\end{bmatrix}, \begin{bmatrix}-1\\1\\-2\end{bmatrix}\right\}$$

is linearly independent since the two vectors are not multiples of each other, and thus this set is a basis for W. Thus $\dim(W) = 2$ and W is a plane.

(b) We use the Gram–Schmidt procedure. Set

$$\mathbf{x}_1 = \begin{bmatrix}2\\1\\1\end{bmatrix} \qquad \mathbf{x}_2 = \begin{bmatrix}-1\\1\\-2\end{bmatrix}$$

to be the original vectors. Now let $\mathbf{u}_1 = \mathbf{x}_1$ and

$$\mathbf{u}_2 = \mathbf{x}_2 - \left(\frac{\mathbf{x}_1 \cdot \mathbf{u}_1}{\mathbf{u}_1 \cdot \mathbf{u}_1}\right)\mathbf{u}_1 = \begin{bmatrix}-1\\1\\-2\end{bmatrix} - \frac{-3}{6}\begin{bmatrix}2\\1\\1\end{bmatrix} = \begin{bmatrix}0\\\frac{3}{2}\\-\frac{3}{2}\end{bmatrix}.$$

An orthogonal basis for W is $\{\mathbf{u}_1, \mathbf{u}_2\}$.

(c) *Important: you have to use an orthogonal basis for this.*

The formula for the projection is

$$\hat{\mathbf{v}} = \left(\frac{\mathbf{v} \cdot \mathbf{u}_1}{\mathbf{u}_1 \cdot \mathbf{u}_1}\right)\mathbf{u}_1 + \left(\frac{\mathbf{v} \cdot \mathbf{u}_2}{\mathbf{u}_2 \cdot \mathbf{u}_2}\right)\mathbf{u}_2 = \frac{2}{3}\begin{bmatrix}2\\1\\1\end{bmatrix} + \frac{2}{3}\begin{bmatrix}0\\\frac{3}{2}\\-\frac{3}{2}\end{bmatrix} = \begin{bmatrix}\frac{4}{3}\\\frac{5}{3}\\-\frac{1}{3}\end{bmatrix}.$$

(d) The distance is the magnitude of the orthogonal component of $\mathbf{v}$ with respect to W:

$$\|\mathbf{v} - \hat{\mathbf{v}}\| = \left\| \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} - \begin{bmatrix} \frac{4}{3} \\ \frac{5}{3} \\ -\frac{1}{3} \end{bmatrix} \right\| = \left\| \begin{bmatrix} -\frac{1}{3} \\ \frac{1}{3} \\ \frac{1}{3} \end{bmatrix} \right\| = \sqrt{\frac{1}{3}} = \frac{1}{\sqrt{3}}$$

Note: It is a good idea to check your work by checking the vector $\mathbf{v} - \hat{\mathbf{v}}$ from above is in fact orthogonal to W.

(e) We need a basis for the null space of

$$\begin{bmatrix} 2 & 1 & 1 \\ -1 & 1 & -2 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \end{bmatrix}$$

The solution is

$$W^{\perp} = \operatorname{span} \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}.$$

(f) Since we now know that every vector in

$$\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$$

in W is orthogonal to

$$\begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix},$$

this means W is the solution set for the equation

$$\begin{bmatrix} -1 & 1 & 1 \end{bmatrix} \mathbf{x} = 0.$$

Problem 9. Consider the map $D : \mathbb{P}_2 \to \mathbb{P}_2$ given by $D(p(t)) = p'(t)$, where $p'(t)$ is the calculus derivative of the polynomial $p(t)$. Recall that this map is a linear transformation.

(a) Compute $D(a_0 + a_1 t + a_2 t^2)$.

(b) Find the matrix for D with respect to the standard basis $\mathcal{B} = \{1, t, t^2\}$ for $\mathbb{P}_2$.

(c) Is D injective? Is D surjective? Explain your answer.

Answer 9.

(a) $D(a_0 + a_1 t + a_2 t^2) = a_1 + 2a_2 t$.

(b) We just need to find the images of $1, t, t^2$ under D and write their coordinate vectors in the basis $\mathcal{B}$.

$$D(1) = 0 \implies [D(1)]_{\mathcal{B}} = [0]_{\mathcal{B}} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$D(t) = 1 \implies [D(t)]_{\mathcal{B}} = [1]_{\mathcal{B}} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \qquad \implies A_{\mathcal{B}\leftarrow\mathcal{B}} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{bmatrix}.$$

$$D(t^2) = 2t \implies [D(t^2)]_{\mathcal{B}} = [2t]_{\mathcal{B}} = \begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix}.$$

(c) The matrix for D in the basis $\mathcal{B}$ does not have a pivot in every row nor a pivot in every column, so D is neither injective nor surjective.

Problem 10. If possible, diagonalize the matrix

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 3 \end{bmatrix}.$$

If A is not diagonalizable, explain why.

Answer 10. Since A is upper triangular, its eigenvalues are the diagonal entries:

$$\lambda_1 = 1, \quad \lambda_2 = 2, \quad \lambda_3 = 3.$$

Eigenvectors for $\lambda_1 = 1$:

$$A - I = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 2 \end{bmatrix} \sim \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \implies \text{Every} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \in \text{null}(A - I) \text{ has } \begin{cases} x_2 = 0 \\ x_3 = 0 \\ x_1 \text{ free variable} \end{cases}$$

so one eigenvector is

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}.$$

Eigenvectors for $\lambda_2 = 2$:

$$A - 2I = \begin{bmatrix} -1 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & -1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \implies \text{Every} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \in \text{null}(A - 2I) \text{ has } \begin{cases} x_1 = x_2 \\ x_3 = 0 \\ x_2 \text{ free variable} \end{cases}$$

so one eigenvector is

$$\mathbf{v}_2 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}.$$

Eigenvectors for $\lambda_3 = 3$:

$$A - 3I = \begin{bmatrix} -2 & 1 & 0 \\ 0 & -1 & 1 \\ 0 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -\frac{1}{2} \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix} \quad \Longrightarrow \quad \text{Every} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \in \text{null}(A - 3I) \text{ has } \begin{cases} x_1 = \frac{1}{2}x_2 \\ x_3 = x_2 \\ x_2 \text{ free variable} \end{cases}$$

so one eigenvector is

$$\mathbf{v}_3 = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}.$$

Since the sum of the dimensions of the eigenspaces is $1 + 1 + 1 = 3$, A is diagonalizable. So we can take

$$P = \begin{bmatrix} \mathbf{v}_1 & \mathbf{v}_2 & \mathbf{v}_3 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 2 \end{bmatrix} \qquad \text{and} \qquad D = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}.$$

Then $A = PDP^{-1}$.

Problem 11. If possible, diagonalize

$$B = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}.$$

If B is not diagonalizable, explain why.

Answer 11. The matrix is upper triangular, so its eigenvalues are just the values on the diagonal, which are

$$\lambda_1 = 1 \text{ (algebraic multiplicity 2)}, \qquad \lambda_2 = 2 \text{ (algebraic multiplicity 1)}.$$

Let us compute the eigenspace for $\lambda_1 = 1$:

$$B - I = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \sim \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

This matrix has only one free variable (x_1), so the dimension of null($B - I$) is 1. Therefore, the dimension of the eigenspace of B associated to 1 is only 1. But the algebraic multiplicity of $\lambda = 1$ is 2. Since

$$\dim E_1(B) < 2,$$

there are not enough eigenvectors to form a basis of $\mathbb{R}^3$. Therefore, B is <u>not</u> diagonalizable.

Problem 12. Find A^{-1}, where A is a 2×2 matrix such that

$$A \begin{bmatrix} 1 \\ 1 \end{bmatrix} = 5 \begin{bmatrix} 1 \\ 1 \end{bmatrix} \qquad A \begin{bmatrix} 1 \\ -1 \end{bmatrix} = 7 \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

(*Hint:* Use diagonalization.)

Answer 12. The information we are given shows that the eigenvalues of A are 5 and 7, and that

$$\begin{bmatrix}1\\1\end{bmatrix} \text{ is an eigenvector associated to } 5 \quad \text{and} \quad \begin{bmatrix}1\\-1\end{bmatrix} \text{ is an eigenvector associated to } 7.$$

Thus A is diagonalizable, with diagonalization

$$A = \begin{bmatrix}1 & 1\\1 & -1\end{bmatrix}\begin{bmatrix}5 & 0\\0 & 7\end{bmatrix}\begin{bmatrix}1 & 1\\1 & -1\end{bmatrix}^{-1}.$$

Note that

$$A = PDP^{-1} \implies A^{-1} = \left(PDP^{-1}\right)^{-1} = \left(P^{-1}\right)^{-1} D^{-1}P^{-1} = PD^{-1}P^{-1}.$$

Thus

$$A^{-1} = \begin{bmatrix}1 & 1\\1 & -1\end{bmatrix}\begin{bmatrix}\frac{1}{5} & 0\\0 & \frac{1}{7}\end{bmatrix}\begin{bmatrix}1 & 1\\1 & -1\end{bmatrix}^{-1}.$$

Note that

$$\begin{bmatrix}1 & 1\\1 & -1\end{bmatrix}^{-1} = \frac{1}{-1-1}\begin{bmatrix}-1 & -1\\-1 & 1\end{bmatrix} = \frac{1}{2}\begin{bmatrix}1 & 1\\1 & -1\end{bmatrix}.$$

Thus

$$\begin{aligned}
A^{-1} &= \frac{1}{2}\begin{bmatrix}1 & 1\\1 & -1\end{bmatrix}\begin{bmatrix}\frac{1}{5} & 0\\0 & \frac{1}{7}\end{bmatrix}\begin{bmatrix}1 & 1\\1 & -1\end{bmatrix}\\
&= \frac{1}{2}\begin{bmatrix}\frac{1}{5} & \frac{1}{7}\\\frac{1}{5} & -\frac{1}{7}\end{bmatrix}\begin{bmatrix}1 & 1\\1 & -1\end{bmatrix}\\
&= \frac{1}{2\cdot 35}\begin{bmatrix}7 & 5\\7 & -5\end{bmatrix}\begin{bmatrix}1 & 1\\1 & -1\end{bmatrix}\\
&= \frac{1}{2\cdot 35}\begin{bmatrix}12 & 2\\2 & 12\end{bmatrix}\\
&= \frac{1}{35}\begin{bmatrix}6 & 1\\1 & 6\end{bmatrix}
\end{aligned}$$

Alternative solution: By linearity, the first column of A is

$$A\begin{bmatrix}1\\0\end{bmatrix} = A\left(\frac{1}{2}\left(\begin{bmatrix}1\\1\end{bmatrix}+\begin{bmatrix}1\\-1\end{bmatrix}\right)\right) = \frac{1}{2}A\begin{bmatrix}1\\1\end{bmatrix}+\frac{1}{2}A\begin{bmatrix}1\\-1\end{bmatrix} = \frac{1}{2}\left(5\begin{bmatrix}1\\1\end{bmatrix}+7\begin{bmatrix}1\\-1\end{bmatrix}\right) = \frac{1}{2}\begin{bmatrix}5+7\\5-7\end{bmatrix} = \frac{1}{2}\begin{bmatrix}12\\-2\end{bmatrix} = \begin{bmatrix}6\\-1\end{bmatrix}.$$

Similarly, the second column of A is

$$A\begin{bmatrix}0\\1\end{bmatrix} = A\left(\begin{bmatrix}1\\1\end{bmatrix}-\begin{bmatrix}1\\0\end{bmatrix}\right) = A\begin{bmatrix}1\\1\end{bmatrix} - A\begin{bmatrix}1\\0\end{bmatrix} = \begin{bmatrix}5\\5\end{bmatrix}-\begin{bmatrix}6\\-1\end{bmatrix} = \begin{bmatrix}-1\\6\end{bmatrix}.$$

Therefore,

$$A = \begin{bmatrix}6 & -1\\-1 & 6\end{bmatrix}.$$

Thus

$$\det(A) = 6 \cdot 6 - 1 = 35.$$

$$A^{-1} = \frac{1}{35}\begin{bmatrix} 6 & 1 \\ 1 & 6 \end{bmatrix}.$$

Problem 13. Suppose that A is a diagonalizable $n \times n$ matrix and has only 1 and -1 as eigenvalues. Find A^2.

Answer 13. Since A is diagonalizable, we can write

$$A = PDP^{-1}$$

for some invertible matrix P and some diagonal matrix D whose diagonal entries are all 1 (some number of times, say t times, which is the multiplicity of 1 as an eigenvalue of A) and -1 ($n - t$ times, the multiplicity of -1 as an eigenvalue of A). Then

$$A^2 = PD^2P^{-1}.$$

Note that D^2 is the diagonal matrix whose diagonal entries are the squares of the diagonal entries in D. But $1^2 = 1$ and $(-1)^2 = 1$, so no matter what the multiplicities of 1 and -1 were, all the diagonal entries of D^2 are 1. That is, $D^2 = I$. Thus

$$A^2 = PD^2P^{-1} = PIP^{-1} = PP^{-1} = I.$$

Problem 14. Let A be a 3×3 diagonalizable matrix with eigenvalues $\lambda_1 = 1$, $\lambda_2 = 2$, and $\lambda_3 = 3$. Find the determinant of A, and carefully explain your reasoning.

Answer 14. Since A is 3×3 and has three distinct eigenvalues, it must be diagonalizable, so there is some invertible matrix P such that

$$A = PDP^{-1}$$

where

$$D = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}.$$

Thus

$$\det(A) = \det(P)\det(D)\det(P^{-1}).$$

But

$$\det(P^{-1}) = (\det(P))^{-1},$$

so

$$\det(A) = \det(D)\det(P)\det(P^{-1}) = \det(D).$$

But D is diagonal, so its determinant is the product of the entries in the diagonal, and thus

$$\det(A) = \det(D) = 1 \cdot 2 \cdot 3 = 6.$$

Problem 15. Give examples of linear systems of equations having each of the following properties or explain why such a system cannot exist:

(a) A consistent system of 3 equations in 2 variables.

(b) An inconsistent system of 2 equations in 3 variables.

(c) A homogeneous system $Ax = 0$, where A is a 4×6 matrix and the set of solutions to this system is a line.

(d) A linear system of equations for which the solution set is not a subspace.

(e) A matrix A with the property that $\mathrm{col}(A) = \mathrm{null}(A)$.

(f) A 5×5 matrix A with the property that $\mathrm{col}(A) = \mathrm{null}(A)$.

Answer 15.

(a) The system

$$\begin{cases} x_1 = 0 \\ x_2 = 0 \\ x_1 + x_2 = 0 \end{cases}$$

is consistent, since it is a homogeneous system, and all homogeneous systems are consistent.

(b) The system

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

is inconsistent, as the second equation $0 = 1$ is impossible. Equivalently, we may see that the augmented matrix of the system is

$$\begin{bmatrix} 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

which has a pivot on the last column.

(c) This is impossible. The set of solutions to the system is $\mathrm{null}(A)$, so if it is a line, then $\dim(\mathrm{null}(A)) = 1$. By the Rank–Nullity Theorem,

$$\dim(\mathrm{null}(A)) + \dim(\mathrm{col}(A)) = 6 \implies \dim(\mathrm{col}(A)) = 6 - 1 = 5.$$

But the column space of A is a subspace of $\mathbb{R}^4$, so it cannot have dimension 5.

(d) Consider the system

$$\begin{cases} x_1 = 1 \\ x_2 = 0 \end{cases}$$

whose solutions form a line in $\mathbb{R}^2$. Since $(0, 0)$ is not a solution to this system, the solutions do not form a vector space.

Important: Note that the set of solutions to a <u>homogeneous</u> linear system is always a subspace.

(e) Consider the matrix

$$A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}.$$

Then

$$\operatorname{col}(A) = \operatorname{span}\begin{bmatrix} 1 \\ 0 \end{bmatrix} = \operatorname{null}(A).$$

(f) There is no such matrix. By the Rank-Nulity Theorem,

$$\dim(\operatorname{null}(A)) + \dim(\operatorname{col}(A)) = 5.$$

If $\operatorname{null}(A) = \operatorname{col}(A)$, then

$$5 = \dim(\operatorname{null}(A)) + \dim(\operatorname{col}(A)) = 2\dim(\operatorname{col}(A)).$$

But this is impossible, as $\dim(\operatorname{col}(A))$ is an integer.

Problem 16. Let A be a 3×3 matrix with the following properties:

$$\operatorname{rank}(A) = 2 \qquad \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \text{ is an eigenvector of } A \text{ associated to } 2 \qquad A\begin{bmatrix} 0 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} -1 \\ 4 \\ 6 \end{bmatrix}.$$

(a) Find the first column of A.

(b) Find all the eigenvalues of A.

(c) Is A diagonalizable? Explain your answer.

Answer 16.

(a) The first column of A is given by

$$A\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}.$$

Note that

$$\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} - \begin{bmatrix} 0 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

and the information about the eigenvalue 2 gives us

$$A\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = 2\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 2 \\ 4 \\ 6 \end{bmatrix}.$$

Therefore, the first column is

$$A\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = A\left(\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} - \begin{bmatrix} 0 \\ 2 \\ 3 \end{bmatrix}\right) = A\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} - A\begin{bmatrix} 0 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 2 \\ 4 \\ 6 \end{bmatrix} - \begin{bmatrix} -1 \\ 4 \\ 6 \end{bmatrix} = \begin{bmatrix} 3 \\ 0 \\ 0 \end{bmatrix}.$$

(b) We are told that 2 is an eigenvalue of A. Moreover, we are also told that A has rank 2, and by the Rank–Nullity Theorem, we conclude that

$$\dim(\text{null}(A)) = 3 - \text{rank}(A) = 1.$$

Therefore, there are nontrivial solutions to $Ax = 0$, so 0 is an eigenvalue of A. Finally, we saw that

$$A\begin{bmatrix}1\\0\\0\end{bmatrix} = \begin{bmatrix}3\\0\\0\end{bmatrix} = 3\begin{bmatrix}1\\0\\0\end{bmatrix},$$

so 3 is an eigenvalue of A.

(c) Yes, A is diagonalizable, since it is a 3×3 matrix with 3 distinct eigenvalues, and thus each eigenspace has dimension 1, so that A must be diagonalizable.

Problem 17. Let A be a 3×3 matrix that has 2 as an eigenvalue and such that

$$\dim(\text{null}(A^\top)) = 2.$$

Is A diagonalizable? Explain.

Answer 17. Note that $\text{null}(A^\top) = \text{col}(A)^\perp$, so we conclude that

$$\dim(\text{col}(A)^\perp) = 2.$$

Since

$$\dim(\text{col}(A)) + \dim(\text{col}(A)^\perp) = 3,$$

we conclude that $\text{col}(A)$ has dimension 1. By the Rank–Nullity Theorem,

$$\dim(\text{null}(A)) = 3 - \dim(\text{col}(A)) = 3 - 1 = 2.$$

Thus 0 is an eigenvalue of A, and the eigenspace of A associated to 0 has dimension 2. Moreover, we are told that A has another eigenvalue, 2; the only possibility for the dimension of the eigenspace associated to 2 is 1, since the sum of the dimensions of the various eigenspaces is at most 3. In particular, the sum of the dimensions of the eigenspaces of A is $2 + 1 = 3$, so A must be diagonalizable.

Problem 18. Give an example or explain why no example exists.

(a) A 2×2 matrix A with eigenvalues -1, 2, and 3 that is not diagonalizable.

(b) A 3×3 matrix A with eigenvalues -1, 2, and 3.

(c) A 3×3 matrix A with eigenvalues -1, 2, and 3 that is not diagonalizable.

(d) A 4×4 matrix A with eigenvalues -1, 2, and 3 that is not diagonalizable.

(e) A 3×3 matrix A with eigenvalues 2 and 5 that is not diagonalizable.

(f) A 3×3 matrix A with eigenvalues 2 and 5 that is diagonalizable.

(g) A 4×4 matrix A with no (real) eigenvalues.

(h) A 31×31 matrix A with no (real) eigenvalues.

(i) A 314×314 matrix A with $\det(A) = 0$ and no (real) eigenvalues.

(j) An upper-triangular 4×4 matrix A with no (real) eigenvalues.

Answer 18.

(a) A 2×2 matrix A with eigenvalues -1, 2, and 3 that is not diagonalizable.

Impossible: An $n \times n$ matrix can have at most n distinct eigenvalues.

(b) A 3×3 matrix A with eigenvalues -1, 2, and 3.

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}.$$

(c) A 3×3 matrix A with eigenvalues -1, 2, and 3 that is not diagonalizable.

Impossible: an $n \times n$ matrix with n distinct eigenvalues must be diagonalizable.

(d) A 4×4 matrix A with eigenvalues -1, 2, and 3 that is not diagonalizable.

Arrange for one of the eigenvalues λ to have $\operatorname{gemu}(\lambda) = 1 < 2 = \operatorname{almu}(\lambda)$, by the usual trick. Here's an example with $\lambda = -1$:

$$\begin{bmatrix} -1 & 1 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \end{bmatrix}.$$

(e) A 3×3 matrix A with eigenvalues 2 and 5 that is not diagonalizable.

Similar to the previous example:

$$\begin{bmatrix} 2 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 5 \end{bmatrix}.$$

(f) A 3×3 matrix A with eigenvalues 2 and 5 that is diagonalizable.

$$\begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 5 \end{bmatrix}.$$

(g) A 4×4 matrix A with no (real) eigenvalues.

This was done in the Exam 2 practice problems:

$$A = \begin{bmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

You can check that the characteristic polynomial of A is $(\lambda^2 + 1)^2$, which has no real roots.

(h) A 31×31 matrix A with no (real) eigenvalues.

Impossible: the characteristic polynomial will have odd degree and thus at least one real root.

(i) A 314×314 matrix A with $\det(A) = 0$ and no (real) eigenvalues.

Impossible: if $\det(A) = 0$, then 0 is an eigenvalue of A, so A has at least one real eigenvalue.

(j) An upper-triangular 4×4 matrix A with no (real) eigenvalues.

Impossible: the eigenvalues of an upper-triangular matrix are the entries on the diagonal.

Problem 19. For each part below, determine whether the given matrix is diagonalizable (over $\mathbb{R}$) or you don't have enough information.

(a) $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

(b) $\begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 5 & 6 & 7 \\ 0 & 0 & 8 & 9 \\ 0 & 0 & 0 & 10 \end{bmatrix}$

(c) A 2×2 matrix of rank 1.

(d) A 2×2 matrix of rank 2.

(e) A noninvertible 3×3 matrix with exactly one (possibly repeated) real eigenvalue.

Answer 19.

(a) $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

Diagonalizable: the eigenvalues are 1 and -1, which are distinct, so the matrix is diagonalizable.

(b) $\begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 5 & 6 & 7 \\ 0 & 0 & 8 & 9 \\ 0 & 0 & 0 & 10 \end{bmatrix}$

Diagonalizable: the eigenvalues are 1, 5, 8, and 10, which are distinct, so the matrix is diagonalizable.

(c) A 2×2 matrix of rank 1.

Not enough information. For example, the matrix $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ has rank 1 and is diagonalizable, while the matrix $\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$ has rank 1 and is not diagonalizable.

(d) A 2×2 matrix of rank 2.

Not enough information. For example, the matrix $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ has rank 2 and is diagonalizable, while the matrix $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ has rank 2 and is not diagonalizable.

(e) A noninvertible 3×3 matrix with exactly one (possibly repeated) real eigenvalue.

Not enough information. For example, the matrix $\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ has 0 as its only eigenvalue and is diagonalizable, while the matrix $\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$ has 0 as its only eigenvalue and is not diagonalizable.

Problem 20. For which θ is the (2×2) rotation matrix R_θ diagonalizable? (*Hint:* Think about eigenvalues.)

Answer 20. The eigenvalues of R_θ are $\cos(\theta) \pm i \sin(\theta)$, which are real numbers iff $\sin(\theta) = 0$, iff $\theta = k\pi$ for some integer k. So $\theta = 0$ and $\theta = \pi$ give the only two rotation matrices with real eigenvalues; they also happen to be diagonal:

$$R_0 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \qquad R_\pi = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}.$$

Problem 21. For each item, give an example of a *symmetric* 3×3 matrix A with the requested property, or explain why no such example exists.

(a) A has characteristic polynomial $\lambda(\lambda + 1)(\lambda - 1)$.

(b) A has characteristic polynomial $\lambda(\lambda^2 + 1)$.

(c) A has two eigenspaces,

$$\left\{\begin{bmatrix} x \\ y \\ 0 \end{bmatrix} : x, y \in \mathbb{R}\right\} \quad \text{and} \quad \left\{\begin{bmatrix} t \\ t \\ t \end{bmatrix} : t \in \mathbb{R}\right\}.$$

(d) A has characteristic polynomial $(\lambda - 1)^3$ and $\operatorname{gemu}(1) = 2$.

Answer 21.

(a) A has characteristic polynomial $\lambda(\lambda + 1)(\lambda - 1)$.

Sure:

$$A = \begin{bmatrix} 0 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

(b) A has characteristic polynomial $\lambda(\lambda^2 + 1)$.

Impossible: the eigenvalues of a symmetric matrix must be real numbers, but the roots of $\lambda^2 + 1$ are not real.

(c) A has two eigenspaces,

$$\left\{\begin{bmatrix} x \\ y \\ 0 \end{bmatrix} : x, y \in \mathbb{R}\right\} \quad \text{and} \quad \left\{\begin{bmatrix} t \\ t \\ t \end{bmatrix} : t \in \mathbb{R}\right\}.$$

Impossible: eigenvectors of a symmetric matrix associated to distinct eigenvalues are orthogonal, but the vectors in the two eigenspaces given are not orthogonal. For example: $\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ is in the first eigenspace, $\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ is in the second eigenspace, and their dot product is 1, so they are not orthogonal.

(d) A has characteristic polynomial $(\lambda - 1)^3$ and $\operatorname{gemu}(1) = 2$.

Impossible, since a symmetric matrix is (orthogonally) diagonalizable. But the given conditions force $\operatorname{gemu}(1) = 2 < 3 = \operatorname{almu}(1)$.